

Wojciech Trejter

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Summary

AI engineer and researcher working where machine learning meets the physical world — robotic perception, SLAM, computer vision, and world models that transfer from simulation to reality, with published work on interpretability and quantization. MSc Artificial Intelligence (**Cum Laude**, University of Amsterdam), author in *Transactions on Machine Learning Research*, winner of the **European Rover Challenge 2023**, two-time **ICPC** competitor.

Work History

AI Intern **xBerry** *May – August 2026, Wroclaw, Poland*

- Built a **sim-to-real** transfer pipeline closing the gap between Isaac Lab simulation and real robotic hardware.
- **Fine-tuned** world models with LoRA (Low Rank Adaptation), adapting large models within a limited compute budget.
- Architected simulation pipelines for training and evaluating high-fidelity robotic arm tasks in Isaac Lab.

Teaching Assistant **University of Amsterdam** *Sep 2025 – Apr 2026, Amsterdam*

- Led **seminars and tutorials** for the **Computer Vision** and **Reinforcement Learning** courses, turning abstract concepts (convolutional filters, policy gradients) into example-driven explanations.
- Graded coursework and mentored students on assignments, targeting feedback at commonly misunderstood theory.

Software Engineer Intern **IBM CIC** *Feb 2024 – Jun 2024, Groningen*

- Delivered a **full-stack dashboard** (MongoDB, Express, React, Node — **MERN**) end-to-end, from database schema to UI, for a client delivery team.
- Rotated between **Scrum Master** and Developer roles on a bi-weekly sprint cadence, running standups and sprint reviews.

Technical Team Lead **Makercie** *Jan 2023 – Sep 2023, Groningen*

- Took **1st place** at the **European Rover Challenge Remote Edition (2023)**, leading a 5-person software team.
- Built the rover's autonomy stack in Robot Operating System (**ROS**): computer-vision-driven control of a 7DoF robotic arm.

Education

MSc Artificial Intelligence **University of Amsterdam** *Sep 2024 – Aug 2026, Amsterdam*

- **Cum Laude** with **GPA: 8.3/10**.
- **Thesis**: Detecting long term dynamics for SLAM and 3D Gaussian Splatting applications.
- **Courses**: Computer Vision 1&2, Deep Learning 1&2, Machine Learning 1&2, NLP, MLOps and others.

BSc Artificial Intelligence **University of Groningen** *Sep 2021 – Jul 2024, Groningen*

- **Cum Laude** with **GPA: 8.8/10** and 270 ECTS (180 core + 45 Honours + 45 extracurricular).
- Finished with **Honours**, strengthening **team-work** and **interdisciplinary** approaches to problems.
- **Thesis**: *Application of **Multitask Learning Paradigm** on Hydrodynamic Imaging*.
- **Courses**: Probability Theory, Neural Networks, Computer Graphics, Computational Complexity, and others.

Projects

- **GAPC** (Groningen Algorithmic Programming Contest): Organised the contest and designed its problem set.
- **Image Classification for Antiquities**: Image classification using Deep Learning on a dataset obtained from Portable Antiquities Netherlands.
- **Rodina AI Startup**: Co-founder and programmer for a B2B startup creating synthetic image datasets.

Awards

- **European Rover Challenge Remote Edition (2023)**: 1st place after coordinating a team of 5, as a **Technical Lead**.
- **International Collegiate Programming Contest (2022&2023)**: 28th best programming team in Benelux, 58th in Northwestern Europe.
- **Beta Business Days Startup Competition (2024)**: 1st place, following a rapid development of a viable business strategy.

Publications and papers

- Kurek, I., **Trejter, W.**, Frkovic, S., Erdelez, A., [Re] Improving Interpretation Faithfulness for Vision Transformers, *Transactions on Machine Learning Research 2025*.
  - Improving the **replicability** and **reproducibility** compared to the prior publication.
  - Experience with working on current **SOTA AI models**.
  - Utilizing **HPC Cluster** for the computations.
- **Trejter, W.**, Wolf, B. J., Application of Multitask Learning Paradigm on Hydrodynamic Imaging.
  - Deep learning for signal processing.
  - Use and fine-tuning of Residual Networks.

Skills

Programming & Core: Python, C, Java, SQL, Bash, Linux, Object-Oriented Programming, Dynamic Programming, Debugging, Distributed Computing
Deep Learning & NLP: PyTorch, Keras, scikit-learn, Transformers, Hugging Face, BERT, LLMs, Attention Mechanisms, Fine-tuning, Natural Language Processing, Reinforcement Learning, Generative AI, Diffusion Models
Computer Vision & Robotics: OpenCV, SLAM, 3D Gaussian Splatting, Structure from Motion, ROS, Isaac Lab, Isaac Gym
Data & Infrastructure: NumPy, Pandas, Matplotlib, Parquet, Docker, FastAPI, SLURM, HPC, WandB
MLOps & Engineering: Git, GitLab CI/CD, Hyperparameter Tuning, Model Evaluation, Explainable AI, Unit Testing, Code Review
Methodologies: Agile, Scrum, Kanban

Languages

Polish: Native
English: Full Professional Proficiency

Volunteering

Volunteer Firefighter **Poznan, Poland** *2017-2021*

Emergency Response Officer **Groningen, Netherlands** *2022-2024*